

HR Analytics – Employee Attrition Prediction Report

1. Introduction

Employee attrition is one of the biggest challenges faced by organizations because high employee turnover increases recruitment costs, reduces productivity, and affects overall business performance. This project focuses on analyzing HR employee data to identify the major reasons behind employee resignation and predict future attrition using machine learning techniques.

The project combines:

- Exploratory Data Analysis (EDA)
- Machine Learning Models
- SHAP Explainability
- Interactive Power BI Dashboard

The objective was to help HR teams make data-driven decisions to improve employee retention and workforce planning. HR dashboards commonly focus on turnover rate, overtime, promotions, compensation, and department-level insights.

2. Tools & Technologies Used

Tool	Purpose
Python	Data preprocessing and machine learning
Pandas	Data cleaning and manipulation
Seaborn & Matplotlib	Data visualization
Scikit-learn	Logistic Regression & Decision Tree models
SHAP	Model explainability
Power BI	Interactive HR dashboard

3. Dataset Overview

The HR dataset contains employee-related information such as:

- Department
- Job Role

- Salary
- Gender
- Overtime
- Promotion History
- Attrition Status

The dataset was cleaned and prepared before analysis:

- Missing values checked
- Data types corrected
- Encoded categorical variables transformed
- Salary bands created
- Attrition metrics calculated

Preparing HR datasets properly is important for effective workforce analytics and predictive dashboards.

4. Exploratory Data Analysis (EDA)

Several exploratory analyses were performed to identify patterns related to employee attrition.

4.1 Department-wise Attrition

The analysis showed that the Sales department experienced the highest employee attrition compared to other departments.

4.2 Overtime Impact

Employees working overtime were more likely to leave the organization. Overtime is a major factor influencing employee burnout and resignation. HR attrition dashboards often analyze overtime trends to understand workload imbalance.

4.3 Salary Band Analysis

Employees in lower salary bands showed significantly higher attrition rates than employees with higher salaries.

4.4 Promotion Impact

Employees who had not received promotions for long periods demonstrated a greater tendency to resign.

5. Machine Learning Models

Two classification models were built to predict employee attrition.

5.1 Logistic Regression

Logistic Regression was used as the primary classification algorithm because of its strong performance on binary classification problems.

Result:

- Accuracy: ~88%
- Good prediction performance
- Better generalization

5.2 Decision Tree Classifier

A Decision Tree model was also trained to compare performance.

Result:

- Accuracy: ~77%
- Easier interpretability
- Lower accuracy compared to Logistic Regression

Logistic Regression provided better predictive performance for this dataset.

6. Model Evaluation

The models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

The confusion matrix helped identify:

- Correct attrition predictions
- False positives
- False negatives

The Logistic Regression model achieved higher reliability in predicting employee turnover.

7. SHAP Value Analysis

SHAP (SHapley Additive exPlanations) was used to explain model predictions and identify the most influential features affecting attrition.

Important Features Identified:

- Overtime
- Monthly Income
- Years Since Last Promotion
- Department
- Job Role

SHAP analysis improves transparency and interpretability of machine learning predictions, making HR analytics more explainable and actionable.

8. Power BI Dashboard

An interactive HR Analytics Dashboard was created in Power BI to visualize attrition patterns and workforce insights.

Dashboard Features:

- Attrition Rate KPI
- Department-wise Attrition
- Overtime Impact
- Promotion Impact
- Salary Band Analysis
- Interactive Dropdown Slicers
- Department, Gender, and Job Role Filters

HR dashboards commonly use KPI cards, attrition metrics, salary analysis, and interactive filtering for workforce analytics.

9. Key Insights

The project identified several important business insights:

1. Employees working overtime have higher attrition rates.
2. Low salary employees are more likely to resign.
3. Employees with fewer promotions tend to leave the organization.
4. Sales department employees show higher turnover rates.

5. Salary and overtime were among the strongest predictors in SHAP analysis.

These findings align with common HR attrition dashboard practices and workforce analytics studies.

10. Attrition Prevention Suggestions

Based on the analysis, the following recommendations are suggested to reduce employee attrition:

6. Reduce excessive overtime workload.
7. Improve salary structures for lower-income employees.
8. Provide more frequent promotion opportunities.
9. Improve work-life balance policies.
10. Focus employee retention strategies on high-risk departments.
11. Increase employee engagement and satisfaction programs.

Data-driven HR dashboards are widely used to improve retention planning and workforce strategy.

11. Conclusion

This project successfully combined machine learning, explainable AI, and business intelligence techniques to analyze employee attrition. The dashboard and predictive models helped identify key factors influencing employee resignation and provided actionable HR recommendations.

The project demonstrates how analytics can support organizations in reducing turnover, improving employee satisfaction, and making informed HR decisions.